

MATH 3260 - Chapter 4 Notes: Vector Spaces

Feedback-Revised Student Notes

Kennesaw State University - MATH 3260

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These notes are synchronized with the feedback-revised student notebook. They emphasize definitions, theorems, worked mathematics, conceptual connections, and interpretation while omitting notebook-specific code and worksheet instructions.

Chapter overview

1. **Section 4.1:** Vectors in $\mathbb{R}^n$
2. **Section 4.2:** Vector Spaces
3. **Section 4.3:** Subspaces of Vector Spaces
4. **Section 4.4:** Spanning Sets and Linear Independence
5. **Section 4.5:** Basis and Dimension
6. **Section 4.6:** Rank of a Matrix and Systems of Linear Equations
7. **Section 4.7:** Coordinates and Change of Basis

8. Section 4.8: Applications of Vector Spaces

Each section ends with a terminology table, a worksheet, and complete instructor solutions.

Original-example policy

All worked examples and numerical values are original. Personal-interest themes are used selectively, including the San Antonio Spurs, hiking, badminton, driving, travel, Backstreet Boys, and R&B music. Any prices, routes, coordinates, or datasets are fictional teaching models unless explicitly identified otherwise.

Section 4.1 - Vectors in $\mathbb{R}^n$

Learning objectives

Students should be able to:

- represent vectors geometrically as directed line segments;
- add, subtract, and scale vectors in $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^n$;
- interpret vector operations geometrically;
- use vector algebra properties;
- write one vector as a linear combination of other vectors.

Definition: vector in the plane

A vector $\mathbf{v} = (v_1, v_2)$ can be represented by a directed segment from the origin to (v_1, v_2) . The numbers v_1 and v_2 are its components.

Original application: hiking displacement

A hiker travels according to two displacement vectors, measured in kilometers:

$$\mathbf{u} = (4, 3), \quad \mathbf{v} = (-1, 5).$$

The total displacement is

$$\mathbf{u} + \mathbf{v} = (4 - 1, 3 + 5) = (3, 8).$$

The vector $-\mathbf{u} = (-4, -3)$ reverses the first displacement, while $\frac{1}{2}\mathbf{u} = (2, \frac{3}{2})$ has the same direction and half the magnitude.

The parallelogram rule

The vector sum can be visualized as the diagonal of a parallelogram. Equivalently, translate the second vector so that its initial point is the terminal point of the first.

Vector operations in $\mathbb{R}^n$

For $\mathbf{u} = (u_1, \dots, u_n)$, $\mathbf{v} = (v_1, \dots, v_n)$, and scalar c , define $\mathbf{u} + \mathbf{v} = (u_1 + v_1, \dots, u_n + v_n)$ and $c\mathbf{u} = (cu_1, \dots, cu_n)$.

Spurs-inspired vector arithmetic in $\mathbb{R}^3$

Let

$$\mathbf{a} = (9, 1, 20), \quad \mathbf{b} = (21, -3, 4).$$

Then

$$\mathbf{a} + \mathbf{b} = (30, -2, 24),$$

$$2\mathbf{a} = (18, 2, 40),$$

and

$$\mathbf{b} - 2\mathbf{a} = (3, -5, -36).$$

The entries are Spurs-inspired teaching values; they do not represent a single real dataset.

Solving a vector equation in $\mathbb{R}^4$

Let

$$\mathbf{u} = (2, -1, 3, 0), \quad \mathbf{v} = (1, 4, -2, 5), \quad \mathbf{w} = (-1, 2, 1, -3).$$

Solve

$$3(\mathbf{x} + \mathbf{w}) = 2\mathbf{u} - \mathbf{v} + \mathbf{x}.$$

Move the $\mathbf{x}$ terms together:

$$3\mathbf{x} + 3\mathbf{w} = 2\mathbf{u} - \mathbf{v} + \mathbf{x},$$

$$2\mathbf{x} = 2\mathbf{u} - \mathbf{v} - 3\mathbf{w},$$

so

$$\mathbf{x} = \frac{1}{2}(2\mathbf{u} - \mathbf{v} - 3\mathbf{w}).$$

Substitution gives

$$\mathbf{x} = \left(3, -6, \frac{5}{2}, 2\right).$$

An intermediate algebra step can make the computation easier to follow:

$$\frac{1}{2}(2\mathbf{u} - \mathbf{v} - 3\mathbf{w}) = \mathbf{u} - \frac{1}{2}\mathbf{v} - \frac{3}{2}\mathbf{w}.$$

Definition: linear combination

A vector $\mathbf{x}$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_k$ if $\mathbf{x} = c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k$ for scalars $c_1, \dots, c_k$.

Detailed linear-combination example

Let

$$\mathbf{v}_1 = (1, 2, 0), \quad \mathbf{v}_2 = (0, 1, 1), \quad \mathbf{v}_3 = (2, 0, 1),$$

and let $\mathbf{x} = (7, 4, 5)$. We seek c_1, c_2, c_3 such that

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3.$$

Equating components gives

$$\begin{aligned} c_1 + 2c_3 &= 7, \\ 2c_1 + c_2 &= 4, \\ c_2 + c_3 &= 5. \end{aligned}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 7 \\ 2 & 1 & 0 & 4 \\ 0 & 1 & 1 & 5 \end{array} \right].$$

Apply $R_2 \leftarrow R_2 - 2R_1$:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 7 \\ 0 & 1 & -4 & -10 \\ 0 & 1 & 1 & 5 \end{array} \right].$$

Then $R_3 \leftarrow R_3 - R_2$:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 7 \\ 0 & 1 & -4 & -10 \\ 0 & 0 & 5 & 15 \end{array} \right].$$

Thus $c_3 = 3$, $c_2 = 2$, and $c_1 = 1$. Therefore,

$$\boxed{\mathbf{x} = \mathbf{v}_1 + 2\mathbf{v}_2 + 3\mathbf{v}_3}.$$

Common mistakes

Do not confuse a vector with its length. Add vectors componentwise, not by adding lengths. A linear combination problem is a system for unknown coefficients.

Closing note

Vector operations are the computational language of the rest of the chapter. Spanning, independence, basis, coordinates, and rank all ask increasingly structural questions about linear combinations.

Definitions and terminology

Term	Meaning
Vector	Ordered list of components; geometrically, a directed displacement.
Component	One coordinate of a vector.
Zero vector	Vector whose components are all zero.
Vector addition	Componentwise addition.
Scalar multiplication	Multiplication of every component by the same scalar.
Additive inverse	$-\mathbf{v}$, satisfying $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.
Linear combination	Sum $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$.

Section 4.2 - Vector Spaces

Learning objectives

Students should be able to:

- state and interpret the ten vector-space axioms;
- recognize standard vector spaces such as $\mathbb{R}^n$, P_n , $M_{m,n}$, and spaces of functions;
- verify selected axioms for an unfamiliar set and operations;
- disprove vector-space status efficiently by finding one failed axiom.

Definition: vector space

A vector space is a set V equipped with vector addition and scalar multiplication satisfying closure, associativity, commutativity of addition, a zero vector, additive inverses, two distributive laws, compatibility of scalar multiplication, and scalar identity.

The ten axioms

For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and scalars c, d :

1. $\mathbf{u} + \mathbf{v} \in V$.
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
4. There is $\mathbf{0} \in V$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.
5. For every $\mathbf{u}$ there is $-\mathbf{u}$ with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
6. $c\mathbf{u} \in V$.
7. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
8. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$.
9. $c(d\mathbf{u}) = (cd)\mathbf{u}$.
10. $1\mathbf{u} = \mathbf{u}$.

Important standard vector spaces

- $\mathbb{R}^n$: all real n -tuples;
- P_n : all polynomials of degree at most n , together with the zero polynomial;
- $M_{m,n}$: all $m \times n$ real matrices;
- $C[a, b]$: continuous real-valued functions on $[a, b]$;
- solution sets of homogeneous linear systems.

Example: $M_{2,3}$ is a vector space

If A and B are 2×3 matrices, then $A + B$ is also 2×3 . If c is a scalar, then cA is also 2×3 . Thus closure holds. The remaining axioms follow from ordinary real-number arithmetic applied entrywise. The zero vector is the zero matrix and the additive inverse of A is $-A$.

What is being checked?

We are verifying the vector-space axioms, not merely listing matrix operations. In particular, addition and scalar multiplication remain in $M_{2,3}$, the zero matrix belongs to the space, and every A has an additive inverse $-A$ in the same space.

Example: P_2 is a vector space

A polynomial in P_2 has the form

$$p(x) = a_0 + a_1x + a_2x^2.$$

The sum of two such polynomials and any scalar multiple remain of degree at most two. The zero polynomial belongs to P_2 , and $-p(x)$ is also in P_2 . Therefore P_2 is a vector space under ordinary polynomial operations.

Fast disproof strategy

To show that a set is not a vector space, one counterexample is sufficient. Common targets are failure to contain the zero vector, failure of closure under addition, or failure of closure under multiplication by a negative scalar.

Non-example: polynomials of exactly degree two

Let W be the set of all polynomials of exactly degree two. Consider

$$p(x) = x^2 + x, \quad q(x) = -x^2 + 2.$$

Both are in W , but

$$p(x) + q(x) = x + 2,$$

which has degree one. Thus W is not closed under addition and is not a vector space.

Non-example: a positivity restriction

Let

$$W = \{(x, y) \in \mathbb{R}^2 : x > 0\}.$$

The vector $(1, 0)$ lies in W , but

$$-1(1, 0) = (-1, 0) \notin W.$$

So W is not closed under scalar multiplication. It also does not contain the zero vector.

Optional example: unusual operations

Let $V = \mathbb{R}$ with operations

$$u \oplus v = u + v - 1,$$

and

$$c \odot u = cu - c + 1.$$

The zero vector for $\oplus$ is 1, not 0, because $u \oplus 1 = u$. This example illustrates that the axioms concern the specified operations, not necessarily the usual ones.

Closing note

A vector space is an algebraic environment in which linear combinations behave predictably. The elements need not be arrows; they may be matrices, polynomials, functions, signals, or data objects.

Definitions and terminology

Term	Meaning
Vector space	Set with addition and scalar multiplication satisfying ten axioms.
Closure	An operation keeps results inside the set.
Additive identity	The zero vector of the space.
Additive inverse	Vector that sums with a given vector to the zero vector.
P_n	Polynomials of degree at most n .
$M_{m,n}$	Real $m \times n$ matrices.
$C[a, b]$	Continuous functions on $[a, b]$.

Section 4.3 - Subspaces of Vector Spaces

Learning objectives

Students should be able to:

- apply the subspace test;
- use the zero-vector test to reject affine sets quickly;
- recognize lines and planes through the origin as subspaces;
- analyze subspaces of $\mathbb{R}^n$, polynomial spaces, and matrix spaces;
- understand intersections of subspaces.

Definition and subspace test

Let $W \subseteq V$. Then W is a **subspace of the parent space** V when:

1. the zero vector of V is in W ;
2. $\mathbf{u}, \mathbf{v} \in W$ implies $\mathbf{u} + \mathbf{v} \in W$;
3. $c\mathbf{u} \in W$ for every scalar c .

All three conditions must hold.

Definition: subspace

A nonempty subset W of a vector space V is a subspace when it is itself a vector space using the same addition and scalar multiplication as V .

Subspace test

A nonempty subset $W \subseteq V$ is a subspace if and only if it is closed under addition and scalar multiplication. Equivalently, for all $\mathbf{u}, \mathbf{v} \in W$ and scalars a, b , the combination $a\mathbf{u} + b\mathbf{v}$ lies in W .

Original example: a plane through the origin

Let

$$W = \{(x, y, z) \in \mathbb{R}^3 : x - 2y + z = 0\}.$$

The zero vector belongs to W . If $\mathbf{u}, \mathbf{v} \in W$, then

$$u_1 - 2u_2 + u_3 = 0, \quad v_1 - 2v_2 + v_3 = 0.$$

For scalars a, b ,

$$(a\mathbf{u} + b\mathbf{v})_1 - 2(a\mathbf{u} + b\mathbf{v})_2 + (a\mathbf{u} + b\mathbf{v})_3 = a(0) + b(0) = 0.$$

Hence W is a subspace.

Solving for x gives

$$x = 2y - z.$$

Let $y = s$ and $z = t$. Then

$$(x, y, z) = s(2, 1, 0) + t(-1, 0, 1).$$

Non-example: a parallel affine plane

Consider

$$U = \{(x, y, z) : x - 2y + z = 1\}.$$

The zero vector does not satisfy the equation. Therefore U is not a subspace. Geometrically, it is parallel to W but shifted away from the origin.

Singular matrices are not a subspace

Let S be the set of singular 2×2 matrices. The matrices

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

are singular, but

$$A + B = I_2$$

is invertible. Thus S is not closed under addition.

Intersection of two subspaces

Let

$$W_1 = \{(x, y, z) : x + y + z = 0\},$$

and

$$W_2 = \{(x, y, z) : x - z = 0\}.$$

Their intersection satisfies both equations. Since $x = z$, the first equation gives $2x + y = 0$, so $y = -2x$. Therefore,

$$W_1 \cap W_2 = \{t(1, -2, 1) : t \in \mathbb{R}\},$$

which is a line through the origin and hence a subspace.

A polynomial subspace

The set

$$W = \{p \in P_3 : p(1) = 0\}$$

is a subspace. If $p(1) = q(1) = 0$, then

$$(p + q)(1) = p(1) + q(1) = 0,$$

and

$$(cp)(1) = c p(1) = 0.$$

Every polynomial in W is divisible by $x - 1$.

Common mistake

Containing the zero vector is necessary but not sufficient. After the zero-vector check, closure still must be established.

Closing note

Subspaces are the linear subsets of a vector space. Homogeneous equations naturally produce subspaces because zero right-hand sides are preserved by linear combinations.

Definitions and terminology

Term	Meaning
Subspace	Subset that is a vector space under inherited operations.
Subspace test	Check nonemptiness, closure under addition, and closure under scalar multiplication.
Affine set	Shifted linear set; generally not a subspace unless it passes through the origin.
Intersection	Vectors belonging to both subspaces.
Homogeneous constraint	Linear condition with zero right-hand side.

Section 4.4 - Spanning Sets and Linear Independence

Learning objectives

Students should be able to:

- determine whether a vector is a linear combination of a set;
- describe and visualize spans;
- determine whether a set spans a vector space;
- test linear independence by solving a homogeneous system;
- identify explicit dependence relations.

Definition: span

The span of $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the set of all linear combinations $c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k$. It is the smallest subspace containing S .

Linear-combination example revisited

For

$$\mathbf{v}_1 = (1, 2, 0), \quad \mathbf{v}_2 = (0, 1, 1), \quad \mathbf{v}_3 = (2, 0, 1),$$

we found

$$(7, 4, 5) = \mathbf{v}_1 + 2\mathbf{v}_2 + 3\mathbf{v}_3.$$

Therefore $(7, 4, 5)$ belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

Does the set span $\mathbb{R}^3$?

Place the vectors as columns:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Because

$$\det(A) = 5 \neq 0,$$

the system $A\mathbf{c} = \mathbf{x}$ has a unique solution for every $\mathbf{x} \in \mathbb{R}^3$. Hence the three vectors span $\mathbb{R}^3$.

To span $\mathbb{R}^3$ means every target $\mathbf{b} \in \mathbb{R}^3$ can be written as a linear combination of the given vectors. Equivalently, $A\mathbf{c} = \mathbf{b}$ must be solvable for every $\mathbf{b}$.

Definition: linear independence

The set $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly independent if $c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k = \mathbf{0}$ has only the trivial solution. Otherwise the set is linearly dependent.

Original dependent set in $\mathbb{R}^3$

Let

$$\mathbf{a} = (1, 2, 3), \quad \mathbf{b} = (2, 1, 0), \quad \mathbf{c} = (5, 4, 3).$$

Observe that

$$\mathbf{c} = \mathbf{a} + 2\mathbf{b}.$$

Therefore,

$$\mathbf{a} + 2\mathbf{b} - \mathbf{c} = \mathbf{0}$$

is a nontrivial dependence relation. The set is linearly dependent.

Detailed independence test

Consider

$$S = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}.$$

Set

$$c_1(1, 1, 0) + c_2(0, 1, 1) + c_3(1, 0, 1) = \mathbf{0}.$$

This gives

$$\begin{aligned} c_1 + c_3 &= 0, \\ c_1 + c_2 &= 0, \\ c_2 + c_3 &= 0. \end{aligned}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right].$$

After $R_2 \leftarrow R_2 - R_1$,

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right].$$

After $R_3 \leftarrow R_3 - R_2$,

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right].$$

Thus $c_3 = 0$, then $c_2 = 0$, and $c_1 = 0$. The set is linearly independent.

Polynomial dependence

In P_2 , let

$$p_1(x) = 1 + x, \quad p_2(x) = x + x^2, \quad p_3(x) = 1 + 2x + x^2.$$

Since

$$p_3 = p_1 + p_2,$$

we have $p_1 + p_2 - p_3 = 0$. The set is linearly dependent.

Geometry of spans in $\mathbb{R}^3$

- One nonzero vector spans a line through the origin.
- Two independent vectors span a plane through the origin.
- Three independent vectors span all of $\mathbb{R}^3$.
- Adding a vector already in the span does not enlarge the span.

Closing note

Span asks whether vectors provide enough directions; independence asks whether any direction is redundant. A basis will require both.

Definitions and terminology

Term	Meaning
Span	Set of all linear combinations of a set of vectors.
Spanning set	Set whose span equals the whole target space.
Linearly independent	Only the trivial homogeneous coefficient solution exists.
Linearly dependent	A nontrivial coefficient relation produces zero.
Trivial solution	All coefficients equal zero.
Dependence relation	Explicit nontrivial linear combination equal to zero.

Section 4.5 - Basis and Dimension

Learning objectives

Students should be able to:

- test whether a set is a basis;
- use standard bases for $\mathbb{R}^n$, P_n , and $M_{m,n}$;
- explain uniqueness of basis representation;
- find a basis for a subspace;
- compute dimensions and use dimension-based basis tests.

Basis as spanning without redundancy

For P_2 , the standard set $\{1, x, x^2\}$ spans because every polynomial $a+bx+cx^2$ is a linear combination of these three functions. It is independent because

$$c_1 + c_2x + c_3x^2 = 0$$

forces $c_1 = c_2 = c_3 = 0$.

In an n -dimensional space: n independent vectors form a basis; n spanning vectors form a basis; more than n vectors are dependent; fewer than n cannot span the whole space.

Definition: basis

A basis for V is a set that spans V and is linearly independent. A basis is therefore complete without redundancy.

Standard bases

The standard basis for $\mathbb{R}^3$ is

$$\mathbf{e}_1 = (1, 0, 0), \quad \mathbf{e}_2 = (0, 1, 0), \quad \mathbf{e}_3 = (0, 0, 1).$$

The standard basis for P_2 is

$$\{1, x, x^2\}.$$

A standard basis for $M_{2,2}$ is

$$E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad E_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad E_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Original basis example in $\mathbb{R}^3$

Let

$$B = \{(1, 2, 0), (0, 1, 1), (2, 0, 1)\}.$$

The basis matrix is

$$M_B = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Because $\det(M_B) = 5 \neq 0$, the columns are independent and span $\mathbb{R}^3$. Thus B is a basis.

Uniqueness theorem

If B is a basis for V , every vector in V has exactly one representation as a linear combination of the vectors in B .

Extracting a basis from a spanning set

Consider

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\},$$

where

$$\mathbf{v}_1 = (1, 2, 0), \quad \mathbf{v}_2 = (0, 1, 1), \quad \mathbf{v}_3 = (2, 0, 1), \quad \mathbf{v}_4 = (1, 3, 1).$$

Since $\mathbf{v}_4 = \mathbf{v}_1 + \mathbf{v}_2$, it is redundant. The first three vectors form a basis for the same span.

Definition: dimension

If every basis of V contains n vectors, then $\dim(V) = n$. The zero vector space has dimension zero.

Dimension of common spaces

$$\dim(\mathbb{R}^n) = n, \quad \dim(P_n) = n + 1, \quad \dim(M_{m,n}) = mn.$$

Dimension of a parametrized subspace

Let

$$W = \{(a + b, a - b, 2a) : a, b \in \mathbb{R}\}.$$

Rewrite a general vector:

$$(a + b, a - b, 2a) = a(1, 1, 2) + b(1, -1, 0).$$

The two spanning vectors are not scalar multiples, so they are independent. Therefore

$$\dim(W) = 2.$$

Polynomial subspace with a condition

Let

$$W = \{p \in P_3 : p(1) = 0\}.$$

Every $p \in W$ is divisible by $x - 1$. One basis is

$$\{x - 1, x(x - 1), x^2(x - 1)\} = \{x - 1, x^2 - x, x^3 - x^2\}.$$

Thus $\dim(W) = 3$.

Basis tests in an n -dimensional space

If $\dim(V) = n$, then:

- any independent set of exactly n vectors is a basis;
- any spanning set of exactly n vectors is a basis;
- any set with more than n vectors is dependent;
- any set with fewer than n vectors cannot span V .

Closing note

Dimension measures the number of independent directions in a space. A basis provides a minimal coordinate system for those directions.

Definitions and terminology

Term	Meaning
Basis	Linearly independent spanning set.
Standard basis	Natural coordinate basis for a familiar vector space.
Finite-dimensional	Has a basis with finitely many vectors.
Dimension	Number of vectors in any basis.
Basis extraction	Removing redundant vectors while preserving span.
Unique representation	Each vector has exactly one coefficient list relative to a basis.

Section 4.6 - Rank of a Matrix and Systems of Linear Equations

Learning objectives

Students should be able to:

- find bases for row and column spaces;
- compute rank and nullity;
- find a basis for the nullspace;
- apply the rank-nullity theorem;
- express nonhomogeneous solutions as $\mathbf{x}_p + \mathbf{x}_h$;
- interpret consistency using the column space.

For a nonhomogeneous system, the goal is all solutions:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h,$$

where $\mathbf{x}_p$ is one particular solution and $\mathbf{x}_h \in \text{Null}(A)$.

Rank and rank-nullity

$$\text{rank}(A) = \dim(\text{Col}(A))$$

is the number of pivot columns. If A has n columns,

$$\text{rank}(A) + \text{nullity}(A) = n.$$

This balances directions that contribute to the range with directions lost in the nullspace.

Definitions: row and column spaces

For an $m \times n$ matrix A , the row space is the span of the row vectors in $\mathbb{R}^n$. The column space is the span of the column vectors in $\mathbb{R}^m$.

A shared matrix for the section

Let

$$A = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \\ 2 & 1 & 3 & 1 & 3 \\ 1 & 0 & 2 & 2 & 5 \end{bmatrix}.$$

Its reduced row-echelon form is

$$R = \begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix}.$$

The pivot columns are 1, 2, and 4.

Basis for the row space

The nonzero rows of the row-echelon form form a basis for the row space:

$$\{(1, 0, 2, 0, -1), (0, 1, -1, 0, 2), (0, 0, 0, 1, 3)\}.$$

Basis for the column space

Use the pivot positions from R , but select the corresponding columns from the original matrix A :

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}.$$

Do not use the columns of the RREF as a basis for the original column space.

Definition: rank

The rank of a matrix is the common dimension of its row space and column space. Here $\text{rank}(A) = 3$.

Definition: nullspace

The nullspace $N(A)$ is the set of all solutions of $A\mathbf{x} = \mathbf{0}$. It is a subspace of $\mathbb{R}^n$. Its dimension is the nullity.

Detailed nullspace calculation

From the RREF,

$$\begin{aligned} x_1 + 2x_3 - x_5 &= 0, \\ x_2 - x_3 + 2x_5 &= 0, \\ x_4 + 3x_5 &= 0. \end{aligned}$$

Let the free variables be

$$x_3 = s, \quad x_5 = t.$$

Then

$$x_1 = -2s + t, \quad x_2 = s - 2t, \quad x_4 = -3t.$$

Therefore,

$$\mathbf{x} = \begin{bmatrix} -2s + t \\ s - 2t \\ s \\ -3t \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ 0 \\ -3 \\ 1 \end{bmatrix}.$$

A basis for $N(A)$ is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 0 \\ -3 \\ 1 \end{bmatrix} \right\}.$$

Rank-nullity theorem

For an $m \times n$ matrix, $n = \text{rank}(A) + \text{nullity}(A)$. Here $5 = 3 + 2$.

Nonhomogeneous systems: $\mathbf{x}_p + \mathbf{x}_h$

Consider

$$A\mathbf{x} = \mathbf{b}, \quad \mathbf{b} = \begin{bmatrix} 6 \\ 12 \\ 14 \end{bmatrix}.$$

One particular solution is

$$\mathbf{x}_p = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \\ 3 \end{bmatrix},$$

because $A\mathbf{x}_p = \mathbf{b}$. Every solution is

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h,$$

where $\mathbf{x}_h \in N(A)$. Hence

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -1 \\ 3 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ 0 \\ -3 \\ 1 \end{bmatrix}.$$

Consistency and the column space

The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ lies in $\text{Col}(A)$. The coefficients of a solution express $\mathbf{b}$ as a linear combination of the columns of A .

Equivalent conditions for a square matrix

For an $n \times n$ matrix A , the following are equivalent:

- A is invertible;
- $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$;
- $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;
- A is row-equivalent to I_n ;
- $\det(A) \neq 0$;
- $\text{rank}(A) = n$;

- the rows are independent;
- the columns are independent;
- the columns span $\mathbb{R}^n$.

Common mistake

Pivot columns are located using the RREF, but a column-space basis must be taken from the original matrix.

Closing note

Rank counts independent output directions; nullity counts independent input directions that are lost. Together they account for every input coordinate.

Definitions and terminology

Term	Meaning
Row space	Span of matrix rows.
Column space	Span of matrix columns.
Rank	Dimension of row or column space.
Nullspace	Solution space of $A\mathbf{x} = 0$.
Nullity	Dimension of the nullspace.
Pivot column	Column corresponding to a leading entry in echelon form.
Particular solution	One solution of a nonhomogeneous system.
Homogeneous solution	Any vector in the nullspace.
Rank-nullity theorem	$n = \text{rank}(A) + \text{nullity}(A)$.

Section 4.7 - Coordinates and Change of Basis

Learning objectives

Students should be able to:

- find coordinates relative to an ordered basis;
- reconstruct a vector from its coordinates;
- compute transition matrices between bases;
- convert coordinate vectors in both directions;
- represent polynomials and matrices using nonstandard bases.

Ordered bases and the goal of transition matrices

Coordinates require an **ordered basis**

$$B = (\mathbf{b}_1, \dots, \mathbf{b}_n).$$

The transition matrix has the explicit purpose

$$[\mathbf{v}]_C = P_{C \leftarrow B}[\mathbf{v}]_B.$$

Definition: coordinate vector

If $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an ordered basis and $\mathbf{x} = c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n$, then $[\mathbf{x}]_B = (c_1, \dots, c_n)^T$.

Coordinates in a nonstandard basis

Let

$$B = \{(1, 1), (1, -1)\}, \quad \mathbf{x} = (9, 21).$$

We solve

$$c_1(1, 1) + c_2(1, -1) = (9, 21).$$

This gives

$$c_1 + c_2 = 9, \quad c_1 - c_2 = 21.$$

Adding the equations gives $2c_1 = 30$, so $c_1 = 15$. Then $c_2 = -6$. Therefore,

$$[\mathbf{x}]_B = \begin{bmatrix} 15 \\ -6 \end{bmatrix}.$$

Coordinate geometry

The vector itself does not change when the basis changes. Only its coordinate description changes. Relative to the standard basis,

$$[\mathbf{x}]_S = \begin{bmatrix} 9 \\ 21 \end{bmatrix},$$

while relative to B ,

$$[\mathbf{x}]_B = \begin{bmatrix} 15 \\ -6 \end{bmatrix}.$$

Definition: transition matrix

The transition matrix $P_{C \leftarrow B}$ satisfies $[\mathbf{x}]_C = P_{C \leftarrow B}[\mathbf{x}]_B$. Its columns are the C -coordinates of the vectors in B .

Detailed 3×3 change of basis

Let

$$B = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\},$$

and

$$C = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}.$$

The basis matrices are

$$M_B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad M_C = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since

$$\mathbf{x} = M_B[\mathbf{x}]_B = M_C[\mathbf{x}]_C,$$

we obtain

$$[\mathbf{x}]_C = M_C^{-1}M_B[\mathbf{x}]_B.$$

Thus

$$P_{C \leftarrow B} = M_C^{-1}M_B = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Gauss-Jordan construction of the transition matrix

To compute $P_{C \leftarrow B}$ without separately finding M_C^{-1} , row reduce

$$[M_C \mid M_B].$$

Start with

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right].$$

Apply $R_2 \leftarrow R_2 - R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right].$$

Apply $R_1 \leftarrow R_1 - R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right].$$

Apply $R_1 \leftarrow R_1 - R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right].$$

The right block is $P_{C \leftarrow B}$.

Converting a vector

For

$$\mathbf{x} = (9, 20, 21),$$

one finds

$$[\mathbf{x}]_B = \begin{bmatrix} 4 \\ 16 \\ 5 \end{bmatrix}.$$

Then

$$[\mathbf{x}]_C = P_{C \leftarrow B} [\mathbf{x}]_B = \begin{bmatrix} -11 \\ -1 \\ 21 \end{bmatrix}.$$

Coordinates in P_3

Relative to the ordered basis

$$B = \{1, x-1, (x-1)^2, (x-1)^3\},$$

write

$$p(x) = 2 + 3x - x^2 + x^3.$$

Expanding around $x = 1$ gives

$$p(x) = 5 + 4(x-1) + 2(x-1)^2 + (x-1)^3.$$

Therefore,

$$[p]_B = \begin{bmatrix} 5 \\ 4 \\ 2 \\ 1 \end{bmatrix}.$$

Common mistake

A transition matrix depends on direction. Always write the arrow notation $P_{C \leftarrow B}$ before calculating.

Closing note

Coordinates are labels relative to a basis. Change of basis changes the labels, not the underlying vector.

Definitions and terminology

Term	Meaning
Ordered basis	Basis whose vector order is fixed.
Coordinate vector	Coefficients of a vector relative to an ordered basis.
Basis matrix	Matrix whose columns are basis vectors.
Transition matrix	Matrix converting coordinates from one basis to another.
Change of basis	Re-expression of the same vector in a different coordinate system.

Section 4.8 - Applications of Vector Spaces

Learning objectives

Students should be able to:

- verify solutions of linear homogeneous differential equations;
- compute and interpret Wronskians;
- test linear independence of solution functions;
- identify standard conic sections;
- eliminate an xy term by rotation of axes;
- visualize rotated conics computationally.

The vector-space connection is independence: a nonzero Wronskian shows that the solution functions are linearly independent and can serve as a basis for the solution space.

The Chapter 4 connection is **change of basis**: a suitable coordinate change reveals the natural axes of a quadratic form. Later, eigenvectors will give a systematic way to find those directions.

Linear homogeneous differential equations

A linear differential equation of order n has the form

$$y^{(n)} + g_{n-1}(x)y^{(n-1)} + \cdots + g_1(x)y' + g_0(x)y = f(x).$$

It is homogeneous when $f(x) = 0$. The solution set of a linear homogeneous differential equation is a vector space because sums and scalar multiples of solutions remain solutions.

Original example: verifying two solutions

Consider

$$y'' - y = 0.$$

For $y_1 = e^x$,

$$y_1'' - y_1 = e^x - e^x = 0.$$

For $y_2 = e^{-x}$,

$$y_2'' - y_2 = e^{-x} - e^{-x} = 0.$$

Thus both functions are solutions.

Definition: Wronskian

For functions $y_1, \dots, y_n$, the Wronskian is the determinant whose rows contain the functions and their derivatives through order $n-1$. For solutions of an n th-order homogeneous linear equation, a Wronskian not identically zero is equivalent to linear independence.

Wronskian of e^x and e^{-x}

$$W(y_1, y_2) = \det \begin{bmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{bmatrix} = -1 - 1 = -2.$$

Since $W = -2 \neq 0$, the functions are linearly independent. Therefore the general solution is

$$y = C_1 e^x + C_2 e^{-x}.$$

Third-order example

The functions 1, x , and x^2 solve $y^{(3)} = 0$. Their Wronskian is

$$W = \det \begin{bmatrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 0 & 2 \end{bmatrix} = 2.$$

They are linearly independent, so every solution is

$$y = C_1 + C_2 x + C_3 x^2.$$

Standard conic forms

Common conics include:

- circle: $(x - h)^2 + (y - k)^2 = r^2$;
- ellipse: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$;
- hyperbola: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$;
- parabola: $(x - h)^2 = 4p(y - k)$ or $(y - k)^2 = 4p(x - h)$.

Original conic examples

A fictional hiking-path parabola is

$$(x - 2)^2 = -8(y - 1).$$

Its vertex is $(2, 1)$, $4p = -8$, so $p = -2$, and the parabola opens downward.

A fictional travel-region ellipse is

$$\frac{(x + 1)^2}{9} + \frac{(y - 2)^2}{4} = 1.$$

Its center is $(-1, 2)$, with semiaxes 3 and 2.

Rotation of axes

Consider

$$5x^2 - 6xy + 5y^2 = 20.$$

The quadratic-form matrix is

$$Q = \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix}.$$

Its orthonormal eigenvectors point along the rotated axes, with eigenvalues 2 and 8. Using the 45° rotation

$$x = \frac{x' - y'}{\sqrt{2}}, \quad y = \frac{x' + y'}{\sqrt{2}},$$

substitution yields

$$2x'^2 + 8y'^2 = 20.$$

Therefore,

$$\frac{x'^2}{10} + \frac{y'^2}{5/2} = 1,$$

which is an ellipse whose principal axes are rotated relative to the original x - and y -axes.

Closing note

Wronskians translate functional independence into a determinant test. Rotation of axes translates a quadratic equation into coordinates aligned with its natural geometry.

Definitions and terminology

Term	Meaning
Linear differential equation	Differential equation linear in y and its derivatives.
Homogeneous	Right-hand side equals zero.
Wronskian	Determinant built from functions and derivatives.
Conic section	Circle, ellipse, parabola, or hyperbola.
Rotation of axes	Orthogonal coordinate change used to remove an xy term.
Principal axes	Natural axes aligned with a conic's quadratic form.